Predicting EDC mode of action from toxicogenomics data using EDTox

Arindam Ghosh[1], Amirhossein Sakhteman[1], Raghavendra Mysore[1], Einari Niskanen[1], Thomas A. Darde[2], Pierre Daligaux[2], Christophe Chesné[2], Jorma J. Palvimo[1], Vittorio Fortino[1]

[1] Institute of Biomedicine, University of Eastern Finland, Kuopio, Finland

[2] Eurosafe, Saint Grégoire, France

Introduction

Endocrine disrupting chemicals (EDCs) are a class of chemicals that have the potential to alter the normal functioning of the endocrine system by mimicking, blocking, or interfering with the hormones of the system. Even though many of the potential effects of exposure to these compounds are well known, their precise mode of action (MoA) is still not predictable. EDTox offers an easy-to-use web interface (http://edtox.fi/) and standalone R-Shiny application for training classifiers for prioritization of chemicals based on their endocrine disruption potential and identification of the possible MoAs. EDTox utilizes known chemical-gene interactions together with toxicogenomic data driven gene networks to train a binary classifier. The use of generalised linear models (GLM) in EDTox provides a way for identification of the features that are responsible for distinguishing the EDCs from non-EDCs and can be used for predicting the possible MoAs of the EDCs. Currently, EDTox includes classifiers based on toxicogenomics data from Open TG-Gates, DrugMatrix and LINCS that can be used for compiling the ED probability scores for chemicals by entering the list of genes that interact with the chemical. Here, we use gene expression data from HepG2 and HepaRG cell lines exposed to EDCs to train new classifiers and identify their potential MoAs. In particular, we observed that the effects of EDCs are mediated through different pathways under different conditions and that they also vary between doses.

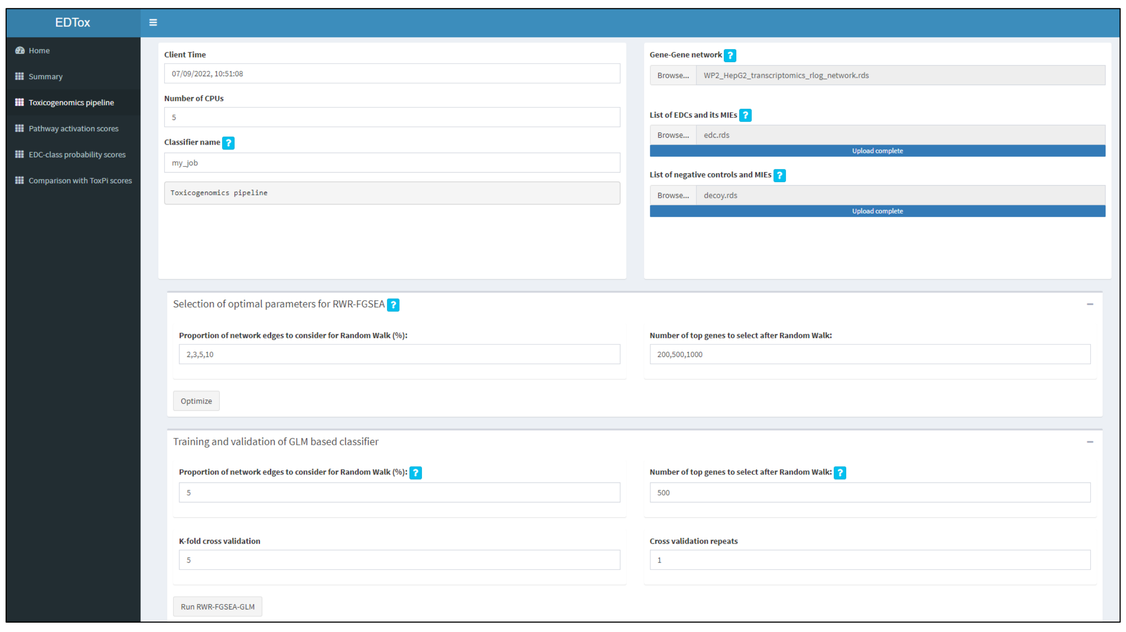


Fig: EDTox tab used for training classifier from toxicogenomic data

Use case data:

EDCs: TPP/TPHP; DEHP; TDCPP; TBT; Carbofuran; 3OH-Carbofuran; PFOS*; Atrazine; PCB 118; PFOA; PCB 153; MEHP; Cypermethrin; DES; Propiconazole; DDE; BPA; Dexamethazone**; GW7647**; GW3965**; CITCO**; GW4064**

Doses: HepG2: variable dose for each EDC; HepaRG: 1nM, 100nM, 10μM

Time points: 24 h in both experiments

[Note: * Not used in HepG2; ** Used only in HepG2]

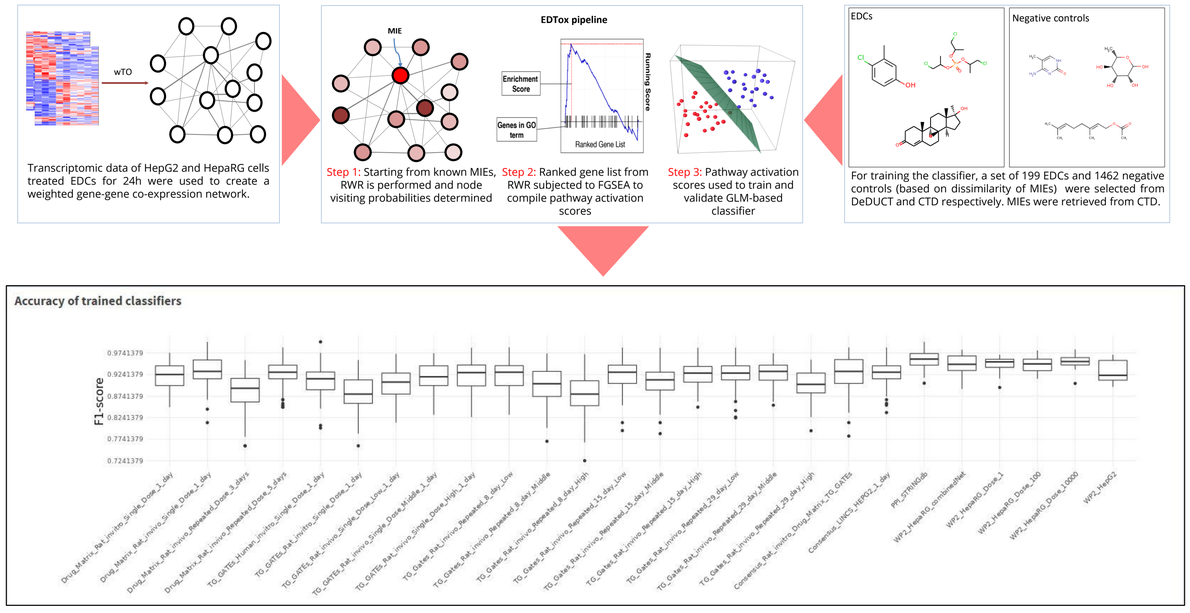
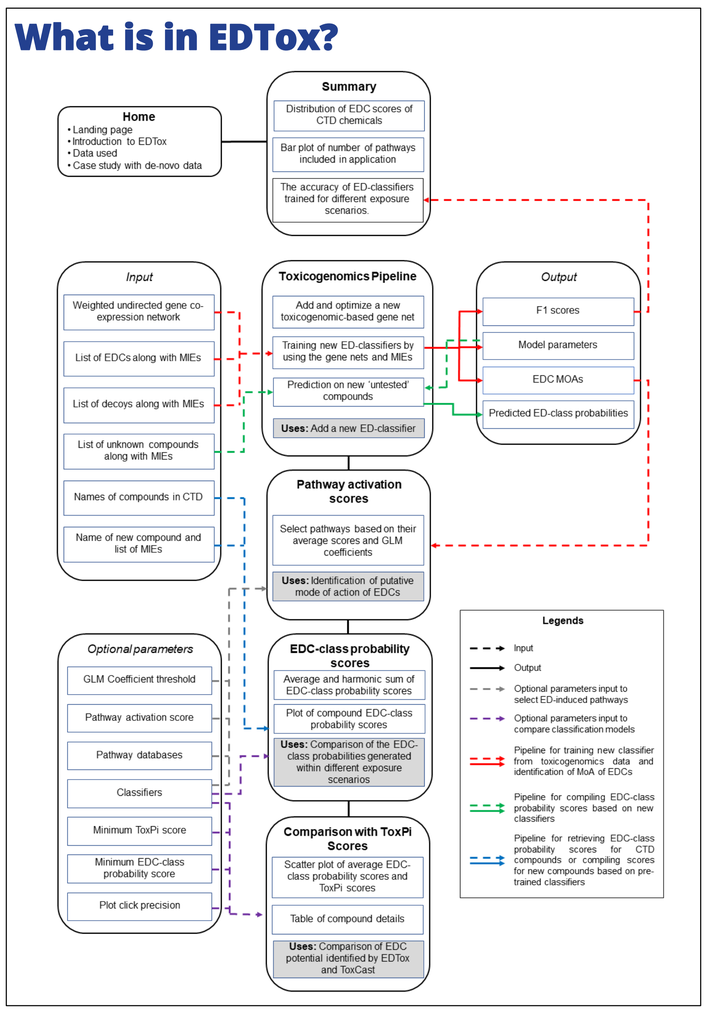


Fig: Comparison of accuracies of the newly trained classifiers with the pre-trained classifiers in EDTox

What is in EDTox?



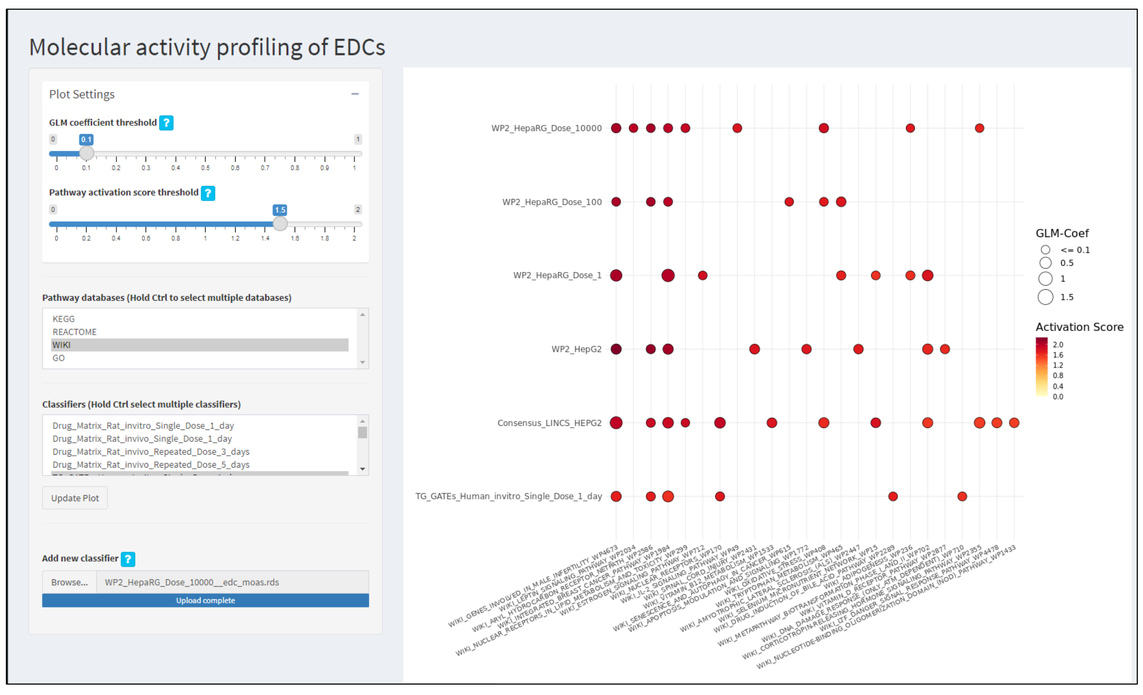


Fig: Systematic comparison of average pathway activation scores and GLM coefficients of pathways for different classifiers. The specific MoA of compounds under different exposure scenarios can be identified by exploring the differences in activated pathways.

Key outcomes

- ✓ Different exposure scenarios exhibit different EDC mode of actions.
- ✓ Yet pathways related to male infertility and breast cancer occurred as the linking MOA across all exposures.
- ✓ Both, male infertility and breast cancer has been previously linked to increased exposure to xenoestrogens.
- ✓ The selection of these two pathways by EDTox can also be linked to the fact that most compounds used for generating the data were xenoestrogens.

CONTACT INFORMATION

vittorio.fortino@uef.fi

arindam.ghosh@uef.fi

MORE INFORMATION

EDTox: edtox.fi

EDCMET: sites.uef.fi/edcmet

BioMedInfoLab: sites.uef.fi/biomedical-informatics

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